

IRUMA AB, Japan

WMO: 476430

Lat: 35.842N

Lon: 139.411E

Elev: 93

StdP: 100.21

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-2.0	-15.0	1.0	4.0	-13.1	1.2	3.8	11.3	6.6	10.2	6.4	1.6	270	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	34.8	25.4	33.1	25.0	31.9	24.6	26.4	31.6	25.9	30.8	25.3	29.9	4.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.4	28.4	24.2	19.4	27.9	24.0	19.1	27.7	82.4	31.8	80.0	31.2	77.9	29.3	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.1	8.6	7.5	DB	-6.0	36.8	1.1	1.1	-6.8	37.6	-7.5	38.3	-8.1	38.9	-8.9	39.7
(5)				WB	-7.6	27.6	0.9	0.9	-8.3	28.3	-8.8	28.8	-9.3	29.3	-10.0	29.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	3.9	4.8	8.1	13.6	18.6	21.8	26.0	27.1	23.2	17.4	11.3	6.2	(6)
		DBStd	8.52	2.15	2.57	3.27	3.53	2.87	2.74	3.19	2.77	3.28	2.97	2.96	2.67	
		HDD10.0	567	191	148	78	9	0	0	0	0	0	19	121		
		HDD18.3	1966	449	380	316	144	32	4	0	3	53	209	375		
		CDD10.0	2477	1	2	21	118	266	354	497	531	395	229	60	4	
		CDD18.3	834	0	0	0	3	39	108	238	273	148	24	0	0	
		CDH23.3	6954	0	0	1	37	254	614	2209	2732	1008	98	1	0	
		CDH26.7	2589	0	0	0	4	46	156	913	1144	309	16	0	0	
(14)	Wind	WSAvg	2.9	3.1	3.4	3.6	3.5	3.1	2.7	2.6	2.7	2.6	2.5	2.5	2.8	(14)
(15)	Precipitation	PrecAvg	1496	52	48	85	104	135	169	170	179	225	192	82	55	(15)
(16)		PrecMax	2088	125	95	159	259	267	394	326	401	405	619	292	160	(16)
(17)		PrecMin	1047	4	2	29	30	57	71	68	42	37	24	3	10	(17)
(18)		PrecStd	209	36	27	34	56	53	65	78	102	110	134	56	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	14.2	17.9	22.0	26.8	30.2	32.9	36.1	36.2	33.8	28.9	21.9	17.1		(19)
(20)		0.4% MCWB	7.5	10.5	12.6	16.3	18.7	23.1	25.7	25.7	24.2	21.1	14.4	11.1		(20)
(21)		2% DB	11.9	14.0	19.0	24.1	27.9	30.1	34.8	35.0	31.9	25.9	19.9	14.8		(21)
(22)		2% MCWB	5.8	7.4	11.1	15.0	17.9	22.0	25.4	25.5	24.0	18.9	13.9	8.5		(22)
(23)		5% DB	10.1	12.0	16.9	22.1	26.0	28.2	33.1	33.8	30.0	24.0	18.1	13.0		(23)
(24)		5% MCWB	4.3	6.0	10.0	14.0	17.1	20.8	25.1	25.3	23.4	17.9	12.5	7.6		(24)
(25)		10% DB	9.0	10.1	14.8	20.1	24.2	26.9	31.8	32.1	28.1	22.2	16.8	11.2		(25)
(26)		10% MCWB	3.6	4.5	8.9	13.4	16.6	20.2	24.8	25.0	22.2	16.9	12.2	6.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	9.2	11.8	14.3	19.2	21.6	25.0	27.2	27.2	26.0	22.4	17.0	12.3		(27)
(28)		0.4% MCDB	11.8	17.5	18.9	23.2	25.6	30.1	32.9	32.6	30.4	26.2	19.6	14.9		(28)
(29)		2% WB	6.8	8.5	12.6	16.9	19.8	23.9	26.3	26.6	25.1	20.6	15.5	10.2		(29)
(30)		2% MCDB	10.3	12.5	17.4	21.5	24.2	27.9	31.7	31.8	29.1	24.1	18.0	12.5		(30)
(31)		5% WB	5.3	6.8	11.0	15.5	18.7	22.7	25.8	26.1	24.4	19.4	14.1	8.6		(31)
(32)		5% MCDB	9.0	10.1	15.5	20.2	23.4	26.3	30.8	31.1	28.1	22.6	16.7	11.6		(32)
(33)		10% WB	4.0	5.3	9.4	14.3	18.0	21.7	25.2	25.6	23.7	18.3	13.0	7.2		(33)
(34)		10% MCDB	8.0	9.0	13.7	18.8	22.0	24.9	29.8	30.3	27.1	21.3	15.3	10.3		(34)
(35)	Mean Daily Temperature Range	MDBR	10.0	9.9	10.2	10.5	9.6	7.7	7.8	8.0	7.4	7.6	8.8	9.7		(35)
(36)		5% DB MCDBR	11.8	12.8	13.8	13.4	12.5	10.4	10.3	10.3	9.6	9.7	10.8	11.3		(36)
(37)		5% WB MCWBR	7.4	8.0	8.1	6.9	5.5	4.4	3.7	3.4	3.9	5.0	6.2	6.9		(37)
(38)		5% WB MCDBR	10.8	11.3	12.3	11.2	10.1	8.1	9.3	9.1	8.4	7.9	8.5	9.5		(38)
(39)		5% WB MCWBR	7.3	7.8	8.1	6.7	5.3	4.0	3.7	3.6	3.9	4.7	5.7	6.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.371	0.419	0.487	0.541	0.570	0.613	0.603	0.597	0.524	0.484	0.417	0.373		(40)
(41)		taud	2.248	2.110	1.950	1.865	1.849	1.812	1.887	1.888	2.049	2.085	2.194	2.274		(41)
(42)		Ebn at Noon	804	802	778	758	742	709	713	706	738	726	742	774		(42)
(43)		Edh at Noon	106	136	175	200	206	214	197	193	157	139	112	98		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.76	3.33	4.14	4.81	5.16	4.55	4.80	4.81	3.79	3.08	2.55	2.41		(44)
(45)		RadStd	0.18	0.29	0.25	0.45	0.54	0.41	0.79	0.47	0.34	0.32	0.23	0.18		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.46	+0.55	N/A	N/A	N/A	N/A	-51	-111	+141	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page